DATE / /

Chapter (1)

أميرة بأخلاقي *

* Database → قاعدة البيانات

A shared collection of related data supporting an organization's activities.

Databases designed
↳ to meet the information needs of an organization.

* Data → بيانات

Collected facts (topic/item) that is unorganized and unstructured (raw facts).

* Information → معلومات

the result of combining, comparing and performing calculations on data.

Data → processing → information

input ↓ Tools ↓ output

الممسوحة ضوئيا بـ CamScanner

DATE / /

Traditional file system concepts

- File Management system also called file system. or file-Based system.
- It's software that handles data file in PC. Manages
- Data storage It stores raw data files on the hard disk. البيانات الاولية
- each file is independent of other file and data in different file. individual program

دمج من طريق

أميرة بأخلاقي

Disadvantages of file system

- why I use data instead of system?
- why they don't use file system in the company?

because

1. Data Redundancy. تكرار البيانات
2. Program-Data Dependency برنامج تبعية البيانات
3. Data Integrity Problems. مشاكل سلامة البيانات
4. Concurrent Access Anomalies. اختلافات الوصول المتزامن

الممسوحة ضوئيا بـ CamScanner

DATE / /

Data Redundancy

→ Data Duplication.
تكرار البيانات

Program-Data ~~Redundancy~~ Dependency

change data file structure $\xrightarrow{\text{مؤدي الى}}$ change application / programs.

أن تغيير هيكل ملف البيانات يؤدي إلى تغيير في برامج/التطبيق التي تصل لملف معين

أهمية بأخلاقي *

Data Integrity Problems

change a data item constraint in a data file

↓ need

change all the application programs.

تغيير قيد عنصر البيانات في ملف البيانات يحتاج إلى التغيير في جميع برامج التطبيقات

Concurrent Access Anomalies

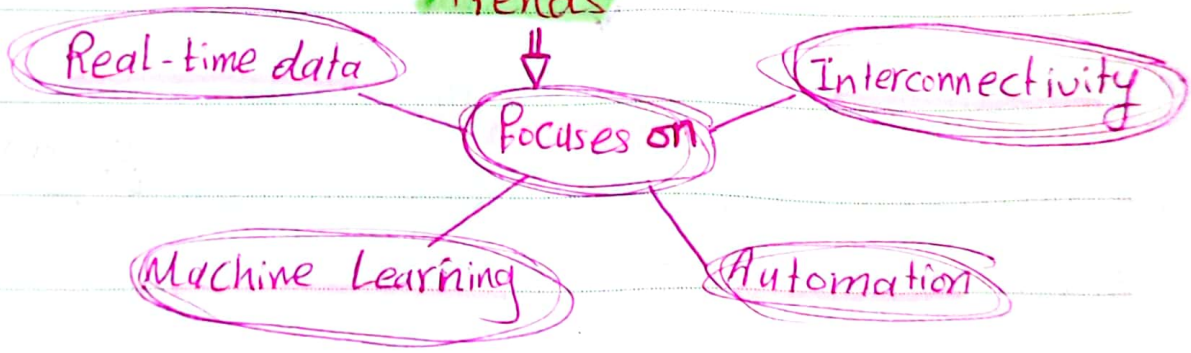
الممسوحة ضوئيا بـ CamScanner

DATE / /

advantages / disadvantages of Traditional File System

(+)	(−)
• simple to use.	• data redundancy.
• less complex.	• data inconsistency.
• Minimal investment. (استثمار)	• Limited data sharing.
• no requirement specialist.	• problem with security.

1.5 Current Technologies and Trends



Referred to

- IoT.
- IIOT.
- Big Data
- Artificial Intelligence.
- Cloud computing.

أميرة بأخلاقي *

الممسوحة ضوئيا بـ CamScanner

DATE / /

IOT (Internet of Things)

connections between physical objects like sensors or machines and the Internet

Example: استخدام شخصي المنزل

- → Home Security
- → Activity Trackers
- → Industrial Security and Safety.
- → Augmented Reality Glasses
- → Motion Detection

IIOT (Industrial Internet of Things)

connections between people and machines as they relate to manufacturing. التصنيع

Example: استخدام للشركات

- → smart city.
- → smart car.
- → Robotics.
- → smart grid.
- → smart communications.

أهمية بأخلاقي *

الممسوحة ضوئيا بـ CamScanner

DATE / /

	IOT	IIOT
Utility	individual customers (homes, offices)	commercial that is industrie (organization, business)
Security	lower	higher
Degree of Application	low-risk impact	high-risk impact
Cost	Less expensive	more expensive

Big Data

Refer to large sets of structured or unstructured data.

أمين بأخلاقي *

that Can be

- compiled تجميعها
- stored تخزينها
- organized تنظيمها
- analysed تحليلها

to reveal الكشف

- patterns → الأنماط
- trends → الاتجاهات
- associations → الجمعيات
- opportunities → الفرص

example: Social Media
Marketing
Heathcare

الممسوحة ضوئيا بـ CamScanner

DATE / /

Artificial Intelligence (AI)

Refers to a computer's ability to perform tasks and make decisions.

Example:

- → Google Maps.
- → Face Detection and Recognition.
- → E-Payment.

Cloud Computing

Practice of using interconnected on the the Internet to → store → manage → and process information.

Example:

- → amazon web services
- → Dropbox
- → Gmail
- → Hotmail

أميرة بأخلاقي *

الممسوحة ضوئيا بـ CamScanner

DATE / /

1.6 Database & DBMS Concept

Database ⇒ is a data storage where information is defined once and can be accessed by different users.

Why do I need a Database solution?

to manage → Multiple Users

→ Multiple components.
or
Multiple data items.

DBMS

أميرة بأخلاقي ✗ :

↓

نظام إدارة قواعد البيانات database management system

↓

is a software that controls the storage organization, and retrieval of data.

الممسوحة ضوئيا بـ CamScanner

DATE / /

أهمية بأخلاقي *

DBMS → software

→ group of programs that

1) access data

2) save data / store data

3) retrieve data

هدفها الأساسي للمستخدم

4) mange data (SQL)

5) orgnazied data

Advantages of database approach

* Reduced Data Redundancy.

تخزين كل عنصر بيانات في مكان واحد فقط في قاعدة البيانات

* Program-Data dependence (low)

File structure is stored in the system catalog, needing only one change to alter the file structure.

* Enforcing Integrity Constraints

have the ability to set rules to make sure users enter vaillid information, maintaining data integrity.

الممسوحة ضوئيا بـ CamScanner

DATE / /

* Concurrent Access

allow many users to access to the same data at the same time.

File System	Database System
easy to use	complex to use
redundancy is high	redundancy is low
inconsistency is high	inconsistency is low
security is low	security is high
stores unstructured data	stores structured data

Example for DBMS?

- Oracle Server اكثر تعقيداً
- MS SQL Server
- My SQL Server

أهمية بالخارجي *

الممسوحة ضوئيا بـ CamScanner

DATE / /

Database Users

- **End Users**

access the database from terminals, using developed applications without knowledge of the database design or working.

- **Database Designers** مرحلة التصميم

Responsible to define content, structure, constraints and functions for the database, communicating with end-users to understand their needs.

- **Application Programmers** مرحلة البرمجة

creat and implement specific programs to access stored data, requiring familiarity with the DBMS for their tasks.

أميرة بأخلاقي *

الممسوحة ضوئيا بـ CamScanner

DATE / /

Database Users

Database Administrators (DBA)

→ single person or a group of person.

→ responsible for everything related to database.

→ responsible for authorizing access to the databas.

for →
- coordinating and monitoring usage.
- controlling use
- monitoring efficiency of operations.

→ responsible for:

- Deciding instances (database).
- Defining Schema.
- user communication.
- security definition. ← يحدد إذا كان تم اختراق قاعدة بيانات
- Backup and recovery.

Backup ↓ نسخ

recovery ↓ استرجاع

أهمية بأخلاق في *

الممسوحة ضوئيا بـ CamScanner